

Using LLM APIs in Google Sheets

Transcription of a Video Tutorial from the GRAPH Network's Generative AI for Work & Research Course

The GRAPH Courses

Introduction

While chat interfaces like ChatGPT and Gemini are common, using Large Language Models (LLMs) through an Application Programming Interface (API) is a powerful skill. An API allows you to call models programmatically using code, which is ideal for automating and repeating tasks hundreds or even thousands of times.

This tutorial will guide you through setting up and using an LLM API directly within Google Sheets to perform data enrichment tasks.

1. Setting Up Your Google Sheet

First, you'll need the data we'll be working with. You can create your own copy of the spreadsheet.

1. **Open the template sheet** (you would typically provide a link here).
2. Select all cells (**Ctrl+A** or **Cmd+A**) and copy them (**Ctrl+C** or **Cmd+C**).
3. Open a brand-new Google Sheet by navigating to **sheets.new** in your browser.
4. Paste the copied data into your new sheet (**Ctrl+V** or **Cmd+V**).

You should now have a sheet with a long list of countries. Our first task will be to find the current **head of state** for each country and then determine their **gender**. While you could copy-paste this into a chat interface, an API is more reliable for this many individual queries.

2. Get an API Key from OpenRouter

We will use **OpenRouter**, a service that provides a unified interface to access many different LLMs.

Sign Up and Add Credits

1. Navigate to openrouter.ai.
2. Sign up for an account (you can use a Google account).
3. After signing in, locate the **Credits** section (usually in the top-right).
4. You'll need to add credits to use the API. The minimum is typically around \$5. Add the funds to your account.
 - *Note: If you are part of a specific program or cohort, you might be provided with a key directly.*

Create and Copy Your API Key

An API key is like a password that grants your code access to your account.

1. Find the **Keys** section on the OpenRouter dashboard (look for a key icon).
 2. Click **Create Key**.
 3. Give your key a descriptive name (e.g., `gpt-sheets-key`).
 4. **Set a credit limit.** This is a crucial safety measure to prevent unexpected costs if your key is ever exposed. A limit of \$3 is more than enough for this exercise.
 5. Click **Create**.
 6. **Immediately copy the generated key.** You will not be able to see it again.
 7. For now, paste this key into a temporary cell in your Google Sheet for safekeeping.
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3. Choose Your Models

OpenRouter provides access to hundreds of models. You can browse them in the **Models** section. For our tasks, we will use two specific models.

1. **Google Gemini 1.5 Flash:** A fast and very cheap model. Its ID is `google/gemini-flash-1.5`.
2. **Perplexity Sonar:** A model optimized for tasks requiring real-time internet search to provide up-to-date information. Its ID is `perplexity/sonar-small-32k-online`.

Copy these two model IDs and paste them into your spreadsheet for easy access later.

4. Create a Custom Function in Google Sheets

To call the API from a sheet formula, we need to create a custom function using Google Apps Script.

1. In your Google Sheet, go to **Extensions > Apps Script**.
2. An editor will open in a new tab. Give your project a name (e.g., “MyGPT”).
3. **Delete** any boilerplate code in the editor (`function myFunction() {...}`).
4. Copy and paste the following JavaScript code into the editor. This function, named MYGPT, will take your prompt, the model ID, and your API key as inputs, send them to OpenRouter, and return the LLM’s response.

```
function MYGPT(prompt, model, key) {
  if (!prompt) return '';
  if (!model) throw new Error('Model ID required (e.g. "openai/gpt-4o").');
  if (!key) throw new Error('OpenRouter API key required.');
```



```
  const url = 'https://openrouter.ai/api/v1/chat/completions';
  const body = {
    model,
    messages: [{ role: 'user', content: String(prompt) }]
  };

  const options = {
    method: 'post',
    contentType: 'application/json',
    headers: { Authorization: 'Bearer ' + key },
    payload: JSON.stringify(body),
    muteHttpExceptions: true
  };

  try {
    const res = UrlFetchApp.fetch(url, options);
    const json = JSON.parse(res.getContentText());
    return json.choices[0].message.content.trim();
  } catch (err) {
    return 'GPT error: ' + err;
  }
}
```

5. Click the **Save project** icon (floppy disk).
6. Click the **Run** button.

Authorize the Script

The first time you run the script, Google will ask for permissions.

1. A “Authorization required” popup will appear. Click **Review permissions**.
 2. Choose the Google account associated with the sheet.
 3. You may see a “Google hasn’t verified this app” screen. This is normal for personal scripts. Click **Advanced**, then click **Go to [Your Project Name] (unsafe)**.
 4. Review the permissions (it needs to connect to an external service, which is the OpenRouter API) and click **Allow**.
 5. You might need to click **Run** one more time in the editor to ensure everything is initialized.
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5. Task 1: Find the Head of State

Now, let’s use our new MYGPT function.

Test the Function

First, test it in any empty cell to make sure it works. The function requires three arguments: `prompt`, `model`, and `key`.

```
=MYGPT("What model are you?", "google/gemini-2.5-flash", "sk-or-...")
```

Replace `"sk-or-..."` with your actual OpenRouter key. After a moment, you should see a response like “I am a large language model trained by Google.”

Create a Dynamic Prompt

To get the head of state for each country, we’ll construct a prompt for each row.

1. Insert a new column to the left of “Head of State” and name it “Prompt”.

2. In cell B2 (assuming countries are in column A), enter the following formula to create a specific prompt for the country in A2:

```
=CONCATENATE("Who is the head of state of ", A2, "?  
Return just their name. If unclear, return 'Undefined'.")
```

3. Drag the fill handle (the small square at the bottom-right of the cell) down to create a prompt for every country.

Apply the API Function

Now, use the MYGPT function to answer each prompt.

1. In the “Head of State” column (cell C2), enter your formula. This time, reference the prompt in cell B2 instead of typing it manually.

```
=MYGPT(B2, "perplexity/sonar", "YOUR_API_KEY_HERE")
```

- **Note:** We are using the **perplexity** model because it can search the web and provide up-to-date information, which is crucial for a topic like current world leaders. The standard Gemini model might have outdated knowledge.
2. After confirming it works for the first row, drag the formula down for all other rows. The sheet will take some time to process all ~200 API calls.

Save Your Results as Values

This is a critical step. Once the formulas have returned all the names, you must convert them from formulas to static text. Otherwise, any small change to the sheet could cause all ~200 API calls to run again, wasting your credits.

1. Select the entire “Head of State” column.
2. Copy it (Ctrl+C).
3. With the column still selected, right-click and choose **Paste special > Values only**.

The cells now contain the plain text names instead of the =MYGPT(...) formula.

6. Task 2: Find the Gender

We will repeat the process to find the gender of each head of state.

1. Create a new “Gender Prompt” column (e.g., column D).
2. In cell D2, create a prompt that uses the country and the leader’s name we just found:

```
=CONCATENATE("What is the gender of ", C2, " (the head of state of ", A2, ")?  
Return only M, F, NB, or Undefined.")
```

3. Drag this formula down for all rows.
4. In the “Gender” column (cell E2), use the MYGPT function again:

```
=MYGPT(D2, "perplexity/sonar", "YOUR_API_KEY_HERE")
```

5. Drag the formula down to get the gender for each person.
6. Once complete, remember to **copy and paste the results as values** to prevent re-running the API calls.

You now have a dataset enriched with up-to-date information, all automated directly within your spreadsheet!